

INTRODUCTION TO STATISTICAL METHODS · COMPANION READER

Sampling and Estimation

How a spoonful tells you about the whole pot: random sampling, the Central Limit Theorem, and confidence intervals

Session 7 (6th–7th June 2026) · Read alongside the lecture slides · Every slide example worked out in full

How to use this companion. Session 6 gave you the moulds of randomness; this session asks the question that makes statistics *useful*: when you can only afford to look at a little of the world, what can you honestly say about all of it? Blue boxes carry the core ideas, amber boxes the analogies, green boxes fully worked slide examples, red boxes the classic traps, purple boxes what to remember on exam day.

1 The soup-tasting problem

★ Everyday picture

A chef does not drink the whole pot to check the seasoning — one stirred spoonful is enough. The pot is the **population**; the spoonful is the **sample**; the *stir* is **randomisation**. Skip the stir and your spoonful might be all chilli from the top — a perfectly measured, beautifully precise, completely wrong answer. Almost everything in this session is about stirring well and knowing how much one spoonful can honestly tell you.

- **Population:** every element of interest — all voters, all phone bills, every unit a factory will ever produce. Measuring it all is a **census**.
- **Sample:** the subset you actually measure — a **sample survey**.
- Why sample at all? The slides put it as 3MT — **Money, Manpower, Material, Time**. Some populations are infinite (all future production) or destroyed by measurement (crash-test cars, taste tests). And for fixed resources, sampling lets a study cover more ground in depth.
- **Inferential statistics:** the machinery for travelling honestly from the spoonful back to the pot.

Parameter vs. statistic: Greek truth, Roman guess

The single most useful notational habit in this course:

Measure	Sample statistic (Roman, computable)	Population parameter (Greek, hidden)
Mean	$\bar{x}$	μ
Variance	s^2	σ^2
Proportion	$\hat{p}$ (or p_s)	P
Correlation	r	ρ

A **parameter** is a fixed property of the population — it exists, but you will typically never see it. A **statistic** is whatever you compute from your sample — visible, but it changes from sample to sample. Statistics are the flashlight beam we point at parameters. Even the distributions of Session 6 were labelled by their parameters: $\text{Binomial}(n, p)$, $\text{Poisson}(\lambda)$, $\text{Normal}(\mu, \sigma^2)$.

✓ Key takeaway

Every observed number carries baggage:

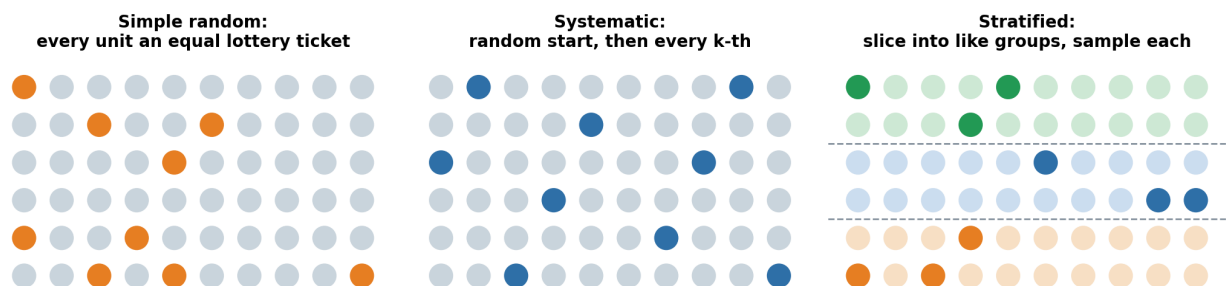
$$\text{Observed data} = \text{Truth} + \text{Bias} + \text{Random error.}$$

Random error is honest noise — it shrinks as samples grow and averages partially cancel it.

Bias is a systematic lean — a sampling method that favours some units — and *no sample size cures it*. A million unstirred spoonfuls still taste of chilli.

2 How to pick a sample

Two families. **Probability sampling:** every population unit has a known, pre-stated chance of selection — this is what makes the mathematics of inference valid. **Non-probability sampling:** units arrive by convenience or judgement, with no probability structure — quick and cheap, but the formulas of this session do not officially apply.



Simple random sampling (SRS). Every unit gets an equal lottery ticket. Requires a homogeneous, finite population and unit-by-unit selection with equal probability. The gold standard against which others are judged.

Systematic sampling. Order the population somehow (alphabetical, by ID, by date of birth), pick a random starting unit, then take every k -th, with $k = N/n$. Operationally easy — think of auditing every 20th invoice. One caution: if the list has a hidden cycle that matches k , you can hit the same kind of unit again and again.

Stratified sampling. When the population is *heterogeneous* but splits into internally similar layers (strata) — hostels, income bands, machine types — sample *within* each layer and combine by weighted average. You guarantee every layer is heard from, which plain SRS cannot promise.

Multistage sampling. Zoom in stages: sample states, then cities within them, then households within those. The practical choice when no full national list exists.

Non-probability cousins. *Convenience* (whoever is nearby — polling your own class), *judgement* (the expert picks “typical” units), *quota* (fill demographic targets by any means), *snowball* (each respondent refers the next — often the only way to reach hidden populations).

✏ Worked example: stratified allocation (slide example)

Three strata of sizes 2000, 6000 and 1000 (total $N = 9000$); a budget of $n = 1500$. Allocate *proportionally to population size*: each stratum gets $n \times N_i/N$:

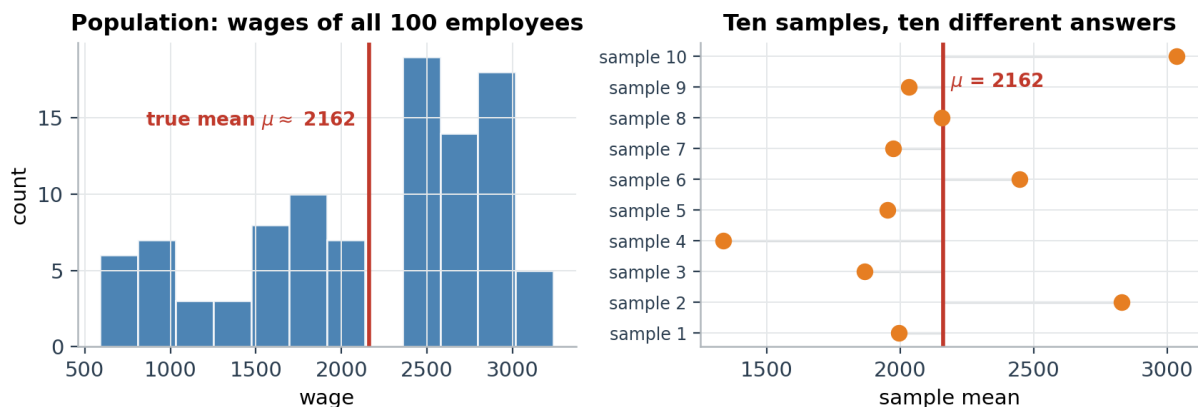
$$2000 \cdot \frac{1500}{9000} = 333, \quad 6000 \cdot \frac{1500}{9000} = 1000, \quad 1000 \cdot \frac{1500}{9000} = 167.$$

The sample becomes a scale model of the population — each layer represented in its true

proportion. Then estimate within each stratum and combine as a weighted average.

3 Sampling variation: every spoonful tastes different

The slides make this concrete with a population of 100 employee wages. Here is that experiment, drawn:



Ten random samples produced ten different means — from 1338 to 3035, scattered around the true $\mu = 2162$. None of the samples *lied*; each told a slightly different part of the truth. This sample-to-sample wobble is **sampling variation**, and three facts about it organise the whole session:

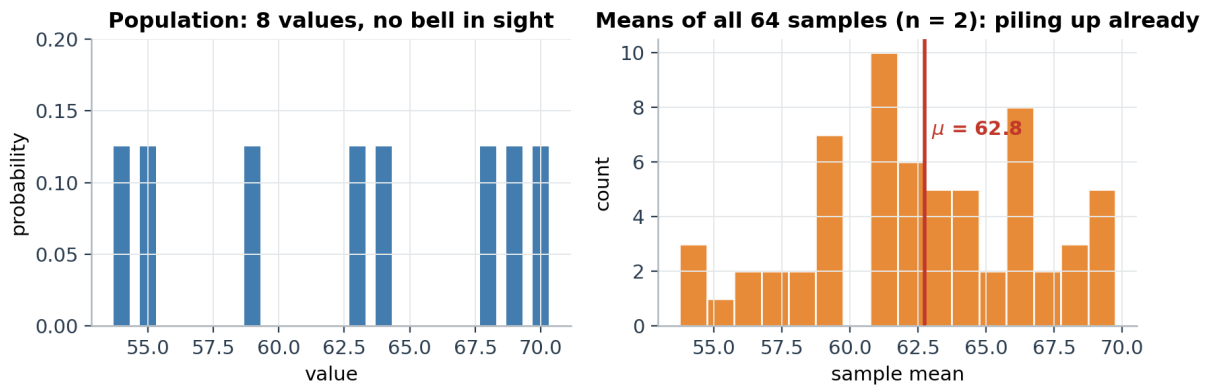
- the statistic (here $\bar{x}$) is itself a *random variable* — it has a distribution;
- the wobble *shrinks* as sample size grows;
- **sampling error** — using an unrepresentative or badly drawn sample — is a different disease from sampling variation, and a much worse one (it is the bias term again).

4 The sampling distribution: the idea the whole course stands on

The intuition

Here is the conceptual leap. You took *one* sample and got *one* $\bar{x}$. Now imagine the parallel universes: every other sample you *could* have drawn, each producing its own $\bar{x}$. The probability distribution of all those possible $\bar{x}$'s is the **sampling distribution of the mean**. You will only ever see one draw from it — but if you know its *shape*, you know exactly how far a typical $\bar{x}$ strays from μ , and that is what licenses every probability statement to come.

The slides build one by brute force. Population: just 8 values {54, 55, 59, 63, 64, 68, 69, 70}, mean $\mu = 62.75$ — a flat, bell-free landscape. Draw every possible sample of size $n = 2$ with replacement: $8 \times 8 = 64$ samples, compute all 64 means.



The population is flat, but the means already *pile up* in the middle and thin out at the ends. Why? A sample mean near 54 requires drawing *both* values small — one way to be extreme, many ways to be middling (a small value paired with a large one lands you in the centre). Averaging manufactures middles. That simple counting fact, scaled up, becomes the most celebrated theorem in statistics.

5 The Central Limit Theorem: order out of chaos

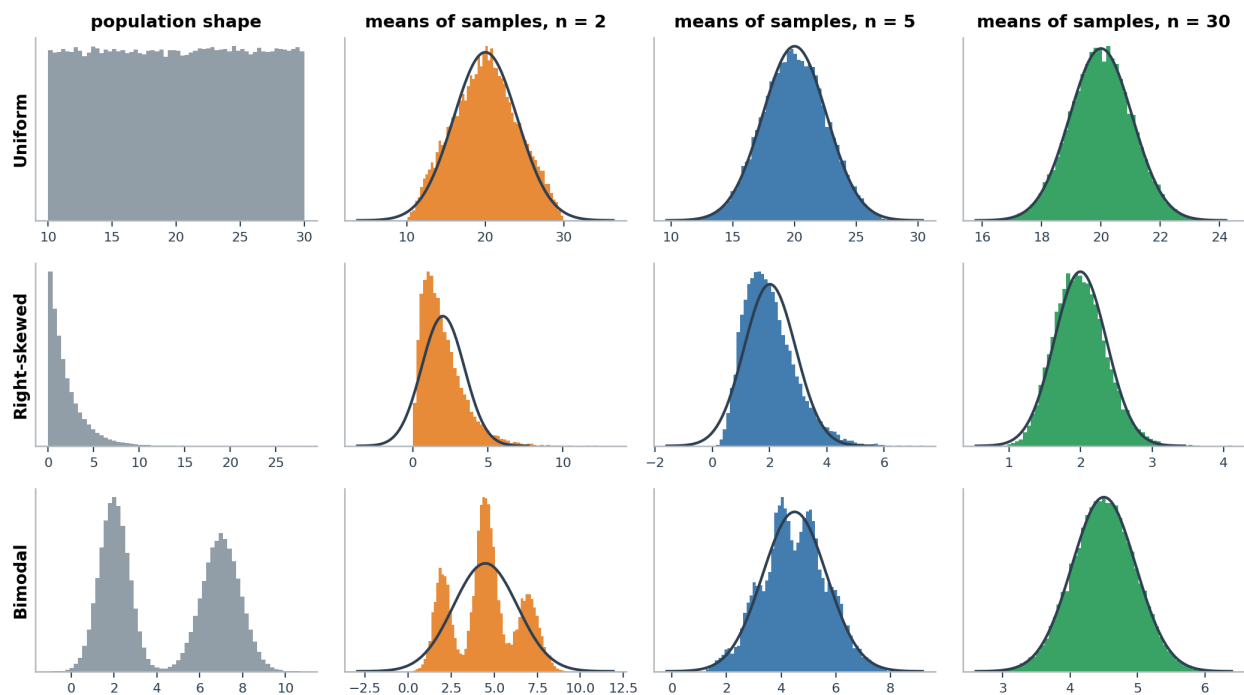
The intuition

Whatever shape the population has — flat, skewed, two-humped, lumpy — **the distribution of sample means becomes normal as n grows**, centring on the true mean with a precisely known spread:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{i.e.} \quad E(\bar{x}) = \mu, \quad SD(\bar{x}) = \frac{\sigma}{\sqrt{n}}.$$

The rule of thumb: $n \geq 30$ is enough for almost any population; if the population is itself normal, *any* n works exactly. The quantity $\sigma / \sqrt{n}$ is so important it has its own name: the **standard error**.

Central Limit Theorem: whatever the population looks like, sample means turn bell-shaped



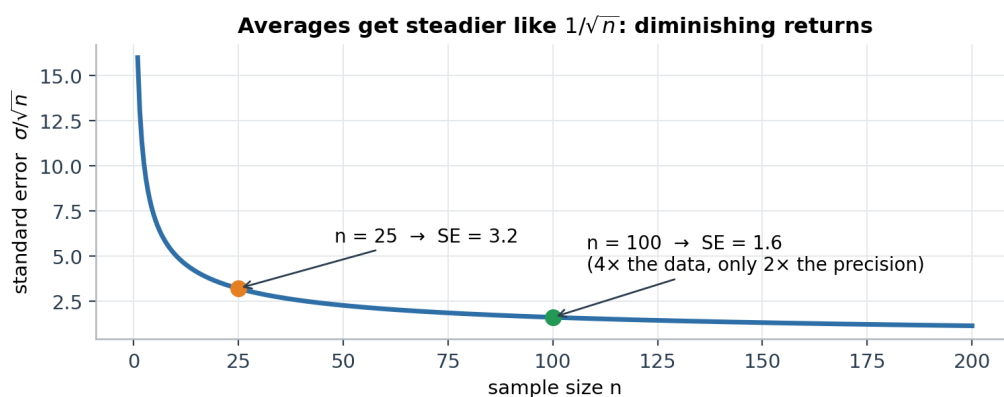
Read the figure row by row: a flat population, a sharply skewed one, a two-humped one. By $n = 30$ all three sampling distributions are clean bells. The population's personality is erased by averaging — only μ and σ survive, as the bell's centre and (after division by $\sqrt{n}$) its width.

★ Everyday picture

Why does averaging create bells? The same reason the dice sums did in Session 6: a sample mean is a *sum of many independent random contributions* (divided by n). Tall and short observations cancel; only conspiracies where *all* draws lean the same way produce extreme means, and conspiracies are rare. The CLT is the mathematical guarantee that this cancellation always wins in the end.

Why $\sqrt{n}$, and what it costs you

The standard error shrinks like $1/\sqrt{n}$, not $1/n$ — precision has *diminishing returns*:



To halve your error you must *quadruple* your sample. This single curve explains half of survey

economics: why polls plateau around 1000–2000 respondents, and why “just collect more data” stops being free advice. (You will see the exact same $4\times$ rule appear in the sample-size problems of Section 10.)

Putting the CLT to work: z-scores for means

Session 6’s ritual carries over with one change — the standard deviation in the denominator is now the standard error:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}.$$

Watch out

The most common exam error in this chapter: using σ where $\sigma / \sqrt{n}$ belongs. Ask yourself: *is the question about one individual, or about a sample average?* “Probability a vehicle is older than 100 months” uses $\sigma = 16$. “Probability the *mean* of 36 vehicles exceeds 100 months” uses $16 / \sqrt{36} = 2.67$. Averages are far steadier than individuals — their bell is $\sqrt{n}$ times narrower.

Worked example: vehicle ages (slide Example 1)

US vehicle ages: $\mu = 96$ months, $\sigma = 16$. For a random sample of $n = 36$, find $P(90 < \bar{x} < 100)$.
Standard error: $\sigma / \sqrt{n} = 16 / 6 = 2.667$.

Convert both ends: $z_1 = (90 - 96) / 2.667 = -2.25$, $z_2 = (100 - 96) / 2.667 = 1.50$.

Assemble: $F(1.50) - F(-2.25) = 0.9332 - 0.0122 = \mathbf{0.921}$.

For one vehicle, the same window (90, 100) would give $z = \mp 0.375$, 0.25 — only about 25% probability. The average of 36 is dramatically more predictable than any single car.

Worked example: tire store spending (slide Example 2)

Spend per customer: $\mu = 85$, $\sigma = 9$; sample $n = 40$. Find $P(\bar{x} \geq 87)$.

$$z = \frac{87 - 85}{9 / \sqrt{40}} = \frac{2}{1.423} = 1.41, \quad P = 1 - F(1.41) = 1 - 0.9207 = \mathbf{0.079}.$$

A \$2 bump sounds small, but in standard-error units it is 1.4 sigmas — unlikely by luck alone. (Foreshadowing: this is exactly the logic hypothesis testing will formalise.)

Worked example: department store shoppers (slide Example 3)

Hourly shoppers: $\mu = 448$, $\sigma = 21$; $n = 49$ hours. Find $P(441 < \bar{x} < 446)$.

Standard error = $21 / 7 = 3$. Then $z_1 = (441 - 448) / 3 = -2.33$ and $z_2 = (446 - 448) / 3 = -0.67$:

$$P = F(-0.67) - F(-2.33) = 0.2514 - 0.0099 = \mathbf{0.2415}.$$

Note the question asks for a window that sits *entirely below* the mean — drawing the bell first makes the subtraction order obvious.

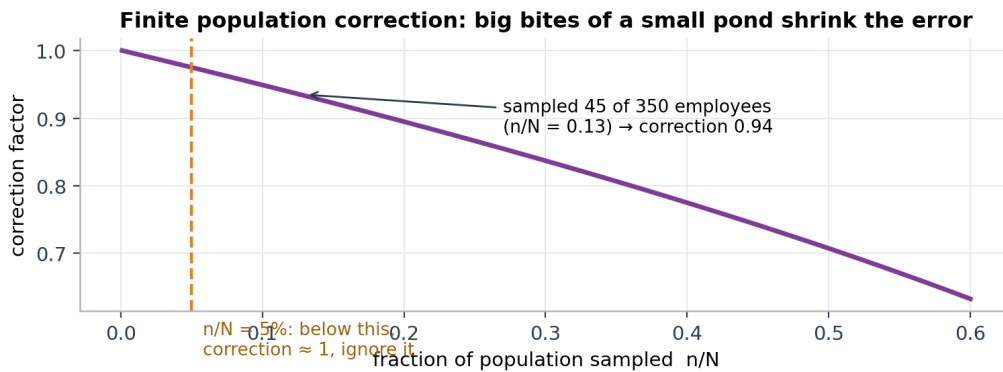
6 Finite populations: big bites from small ponds

The formula $\sigma / \sqrt{n}$ silently assumes the population is effectively infinite — each draw leaves the pond unchanged. But if you sample 45 people from a firm of only 350, each draw meaningfully

shrinks the remaining uncertainty. The correction:

$$SD(\bar{x}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \quad (\text{finite population correction, FPC}).$$

The factor is always ≤ 1 — finite ponds make your estimate *better* than the naive formula claims. Sanity-check the extremes: sample the whole pond ($n = N$) and the factor is 0 — a census has no sampling error at all; sample one unit from millions and the factor is essentially 1.



Rule of thumb from the slides: if $n/N < 5\%$, the factor is so close to 1 that everyone ignores it.

Worked example: hourly employees (slide Example 4)

A company's $N = 350$ hourly employees average $\mu = 37.6$ years with $\sigma = 8.3$. For a sample of $n = 45$, find $P(\bar{x} < 40)$.

Here $n/N = 45/350 = 13\% > 5\%$, so correct:

$$SD(\bar{x}) = \frac{8.3}{\sqrt{45}} \sqrt{\frac{350-45}{349}} = 1.237 \times 0.935 = 1.157.$$

$$z = \frac{40 - 37.6}{1.157} = 2.07, \quad P = F(2.07) = \mathbf{0.9808}.$$

Without the correction, $z = 1.94$ and $P = 0.9738$ — the corrected answer is slightly more confident, exactly as the pond logic predicts.

7 Proportions ride the same train

When the data are yes/no counts — union member or not, defective or not, will vote for her or not — the natural statistic is the **sample proportion**

$$\hat{p} = \frac{x}{n} \quad (\text{e.g. 30 union members of 100 workers} \Rightarrow \hat{p} = 0.30).$$

A proportion is just the *mean of 0/1 Bernoulli data* — so of course the CLT applies to it too:

$$\hat{p} \approx N\left(p, \frac{pq}{n}\right) \quad \text{when } np > 15 \text{ and } nq > 15, \quad z = \frac{\hat{p} - p}{\sqrt{pq/n}}.$$

Everything connects backwards: the pq in the numerator is the Bernoulli variance from Session 6 — maximal at $p = 0.5$, which is why close elections need the largest polls.

 Worked example: calling an election (slide Practice Problem 3, all four parts)

A pollster forecasts a candidate as winner if she gets at least 55% in a sample of $n = 100$.

(a) True support 50.1% (a knife-edge race): $SE = \sqrt{0.501 \times 0.499/100} = 0.050$; $z = (0.55 - 0.501)/0.050 = 0.98$; $P(\hat{p} \geq 0.55) = 1 - F(0.98) = \mathbf{0.164}$. She is barely ahead, and is called the winner only 16% of the time.

(b) True support 60%: $SE = \sqrt{0.24/100} = 0.049$; $z = (0.55 - 0.60)/0.049 = -1.02$; $P = 1 - F(-1.02) = \mathbf{0.846}$. A solid lead is called correctly 85% of the time.

(c) True support 49% (she actually loses): $z = (0.55 - 0.49)/0.050 = 1.20$; $P = 1 - F(1.20) = \mathbf{0.115}$. More than 1 chance in 10 of a wrong call — with $n = 100$, bad nights happen.

(d) Quadruple the sample to $n = 400$: every SE halves, every z doubles: (a) $z = 1.96 \Rightarrow \mathbf{0.025}$; (b) $z = -2.04 \Rightarrow \mathbf{0.979}$; (c) $z = 2.40 \Rightarrow \mathbf{0.008}$. The $\sqrt{n}$ law in action: $4\times$ the respondents, $2\times$ the sharpness — wrong calls in (c) drop from 11.5% to under 1%.

8 Point estimates: one dart at the board

Statistical inference comes in three escalating forms: **point estimation** (one number), **interval estimation** (a range with a confidence level), and **hypothesis testing** (a verdict — next sessions).

A **point estimate** answers “what is μ ?” with a single dart: $\bar{x} = 510$ minutes, $\hat{p} = 0.30$. Its virtues: simple, concrete. Its vice: it is *almost surely wrong* — a continuous dart essentially never lands exactly on the bullseye, and a point estimate is only as good as the representativeness of its sample. Worse, the dart alone does not confess how far off it might be. The wage data made this visible: ten samples, ten darts, scattered from 1338 to 3035.

What we want is a statement that carries its own honesty label — the dart *plus* a circle around it, sized so we can say how often such circles capture the bullseye. That is an interval estimate.

9 Confidence intervals: throwing nets, not spears

★ **Everyday picture**

A point estimate is spear-fishing: precise, and usually a miss. A confidence interval is casting a *net*: “I estimate the mean is 510 minutes, and I am 95% confident it lies between 500 and 520.” Wider nets catch more often; the confidence level is the net-maker’s honest report of the catch rate *of the method*.

Building the net from the CLT

The CLT says $\bar{x}$ lands within 1.96 standard errors of μ for 95% of all samples (that is where the middle 95% of any bell lives). Flip that sentence around: in those same 95% of samples, μ lies within 1.96 standard errors of $\bar{x}$. So the recipe is:

$$\bar{x} \pm \underbrace{z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}_{\text{margin of error } E}$$

$$90\% : z = 1.645$$

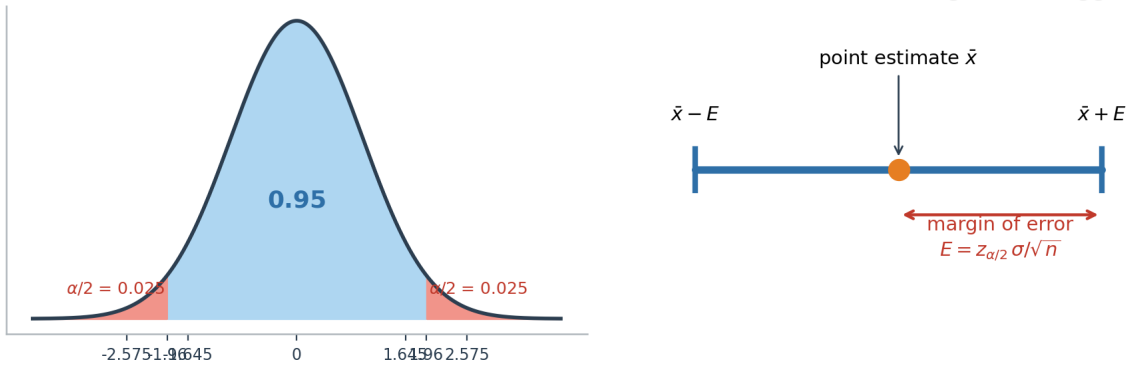
$$95\% : z = 1.96$$

$$98\% : z = 2.33$$

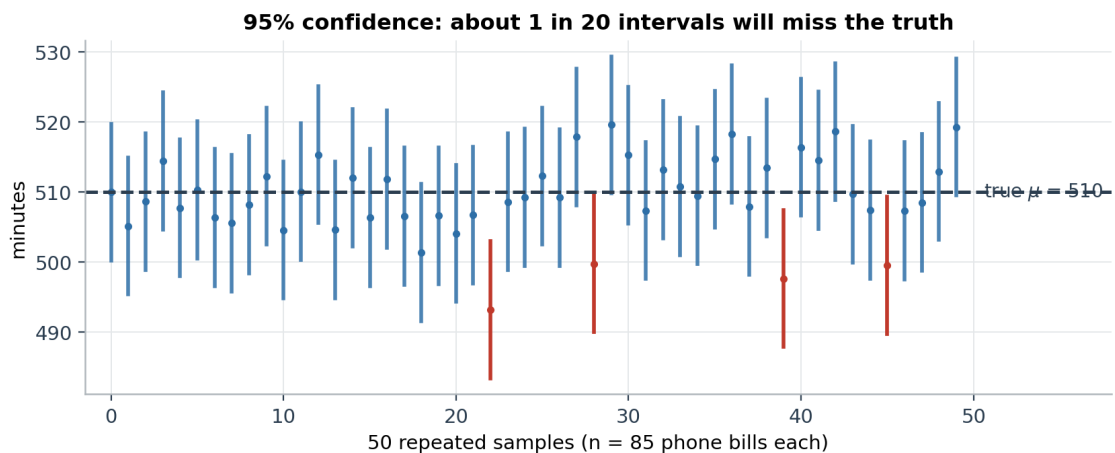
$$99\% : z = 2.575$$

Here α is the *allowed miss rate*: confidence = $1 - \alpha$, split as $\alpha/2$ into each tail of the bell. For 95%, $\alpha = 0.05$ and $z_{0.025} = 1.96$.

Where z-critical values come from (95% shown) A confidence interval = best guess $\pm$ wiggle room



What “95% confident” actually means



⚠ Watch out

“95% confident” does **not** mean “ μ has a 95% chance of being in *my* interval.” Once computed, the interval either contains μ or it does not — μ is a fixed (if hidden) number, not a wanderer. The randomness lives in the *interval*, not the parameter: it is the net that moves from sample to sample, while the fish stays put. The honest sentence: “*the procedure I used captures the true mean in 95% of all samples*” — in the simulation above, 2 or 3 nets out of 50 miss, exactly on schedule. Think ring-toss: the post is fixed; it is your rings that scatter.

📎 Worked example: cellphone minutes (slide example)

From $n = 85$ bills, $\bar{x} = 510$ minutes; history gives $\sigma = 46$. Build the 95% interval.

$$E = 1.96 \times \frac{46}{\sqrt{85}} = 1.96 \times 4.99 = 9.78, \quad 510 \pm 9.78 \Rightarrow [500.2, 519.8] \text{ minutes.}$$

Reported honestly: “we estimate 510 minutes, and with 95% confidence the population mean lies between 500 and 520.”

📎 Worked example: engineers’ ages — 98% and a finite pond (slide example)

$N = 800$ engineers; sample $n = 50$; $\bar{x} = 34.3$ years; $\sigma = 8$; 98% confidence.

Two flags: 98% $\Rightarrow z = 2.33$; and $n/N = 50/800 = 6.25\% > 5\% \Rightarrow \text{FPC}$.

$$E = 2.33 \times \frac{8}{\sqrt{50}} \sqrt{\frac{800-50}{799}} = 2.33 \times 1.131 \times 0.969 = 2.55,$$

$$34.3 \pm 2.55 \Rightarrow [31.8, 36.9] \text{ years.}$$

Demanding more confidence (98% vs 95%) widened the net; the finite pond narrowed it a little. Every term in the formula is a lever you now understand.

Worked example: years trading with India — 90% (slide example)

$n = 44$ companies, $\bar{x} = 10.455$ years, $\sigma = 7.7$, 90% confidence ($z = 1.645$):

$$E = 1.645 \times \frac{7.7}{\sqrt{44}} = 1.645 \times 1.161 = 1.91, \quad 10.455 \pm 1.91 \Rightarrow [8.5, 12.4] \text{ years.}$$

Lower confidence buys a tighter interval — the eternal trade-off. You choose the net width your decision can tolerate.

Two populations, and what if σ is unknown?

Difference of means. To compare two populations (two factories, two cities), the same construction applies to $\bar{x}_1 - \bar{x}_2$; independent variances add, so the standard error becomes

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

A useful reading: if this interval comfortably excludes 0, the two populations plausibly differ — a preview of hypothesis testing.

Unknown σ (the honest case). All examples above were gifted the population σ . In real life you estimate it with the sample s — and Session 6 already told you the price: the statistic $(\bar{x} - \mu) / (s / \sqrt{n})$ follows the *t-distribution* with $n - 1$ degrees of freedom, whose fatter tails demand a slightly larger multiplier. The bounced-cheque problem below uses exactly this.

Worked example: bounced-cheque fees — a t-interval in the wild (slide Practice 4)

Fees from $n = 23$ banks: 26, 28, 20, 20, 21, 22, 25, 25, 18, 25, 15, 20, 18, 20, 25, 25, 22, 30, 30, 30, 15, 20, 29.

From the data: $\bar{x} = 529/23 = 23.0$ and $s = 4.60$. No population σ exists here — only s — so the multiplier comes from t_{22} : for 95%, $t_{22,0.025} = 2.074$ (vs. 1.96 for z — the humility tax).

$$E = 2.074 \times \frac{4.60}{\sqrt{23}} = 1.99 \quad \Rightarrow \quad [21.01, 24.99] \text{ dollars.}$$

Interpretation (part b): we are 95% confident the mean fee across *all* such banks lies between \$21.01 and \$24.99 — a statement about the population mean, not about any single bank's fee.

10 Designing the study: how big a sample?

Until now, n was handed to us. Real studies run backwards: *first* choose the precision you need, *then* solve for the sample size that delivers it. Set the margin of error $E = z\sigma / \sqrt{n}$ to your target and

invert:

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 \quad (\text{always round up — you cannot survey 245.9 people}).$$

 Worked example: the Atlanta restaurant (slide Practice 1)

$n = 49$ diners, $\sigma = \$5$, 95%. **(a)** Margin of error: $E = 1.96 \times 5 / \sqrt{49} = 1.96 \times 0.714 = \1.40 . **(b)** With $\bar{x} = 24.80$: $24.80 \pm 1.40 \Rightarrow [\$23.40, \$26.20]$.

 Worked example: medical expenses — choosing n (slide Practice 5)

Management wants 95% confidence within $E = \$50$; a prior study gives $\sigma \approx \$400$.

(a) $n = (1.96 \times 400 / 50)^2 = (15.68)^2 = 245.9 \Rightarrow \mathbf{246}$ employees.

(b) Twice the precision, $E = \$25$: $n = (31.36)^2 = 983.4 \Rightarrow \mathbf{984}$ — exactly $4 \times$ part (a). The $\sqrt{n}$ law sends its regards: halving the error always quadruples the bill.

11 The one-page cheat sheet

Situation	Distribution / formula	Remember
Sampling distribution of $\bar{x}$	$\bar{x} \sim N(\mu, \sigma^2/n)$	CLT: any population, $n \geq 30$ (any n if population normal)
z for a sample mean	$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$	individuals use σ ; averages use $\sigma / \sqrt{n}$
Finite population	multiply SE by $\sqrt{\frac{N-n}{N-1}}$	only when $n/N > 5\%$
Sample proportion	$\hat{p} \sim N(p, pq/n)$	valid when $np > 15$, $nq > 15$
CI for μ (σ known)	$\bar{x} \pm z_{\alpha/2} \sigma / \sqrt{n}$	z : 1.645 (90%), 1.96 (95%), 2.33 (98%), 2.575 (99%)
CI for μ (σ unknown)	$\bar{x} \pm t_{n-1, \alpha/2} s / \sqrt{n}$	t has fatter tails; s uses the $n - 1$ divisor
CI for $\mu_1 - \mu_2$	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	independent variances add
Sample size for target E	$n = (z_{\alpha/2} \sigma / E)^2$	round up; halving E quadruples n

Glossary in one breath: *population* (the whole pot) · *sample* (the spoonful) · *random sampling* (the stir) · *stratified sampling* (taste every layer) · *parameter* (Greek, hidden, fixed) · *statistic* (Roman, visible, wobbly) · *sampling distribution* (how the statistic wobbles across all possible samples) · *standard error* (the size of that wobble, $\sigma / \sqrt{n}$) · *point estimate* (the dart) · *interval estimate* (the net) · *confidence level* (the net-maker's honest catch rate).

 Key takeaway

The exam-day routine: **(1)** individual or average? — choose σ vs $\sigma / \sqrt{n}$; **(2)** mean or proportion? — SE is $\sigma / \sqrt{n}$ vs $\sqrt{pq/n}$; **(3)** $n/N > 5\%$? — attach the FPC; **(4)** σ known or estimated? — z vs t ; **(5)** for CIs, pick the right $z_{\alpha/2}$ for the stated confidence, and interpret as a property of the *method*, never of one interval; **(6)** sample-size answers always round up.